

NICU Procedures

Pocket guide, procedure edition
Version 3, September 2026

Tags

- 8th NRP 8th edition
- 9th NRP 9th edition (effective 2026)
- Dir. Program director's NNP Brain
- Unit Your unit's poster or order set
- Atlas MacDonald's Atlas of Procedures in Neonatology, 6th ed
- Confirm against your textbook

Sections

- Airway: ETT size, depth, tip, RSI
- Resus: NRP 8th vs 9th, SpO₂, emergency meds
- Lines: UAC, UVC, PICC
- Reference: x-ray targets, unit values, sources

Doses and depths are starting points. Confirm every tube and line clinically and on film, and follow unit protocol.

ETT depth

GA (wk)	Weight (g)	cm
<23 9th	<500	5.0–5.5
23–24	500–600	5.5
25–26	700–800	6.0
27–29	900–1000	6.5
30–32	1100–1400	7.0
33–34	1500–1800	7.5
35–37	1900–2400	8.0
38–40	2500–3100	8.5
41–43	3200–4200	9.0

9th values are measured **at the gum**. 8th used the same bands **at the lip**, confirmed in the Atlas. Atlas

Cross-checks

Nasal-tragus length + 1 cm. Weight (kg) + 6 cm tends to run deep, especially in ELBW infants.

RSI premedication

Order Unit

1	Atropine	0.02 mg/kg IV or IM Atlas
2	Fentanyl	1–4 mcg/kg IV Atlas
3	Rocuronium	0.6–1.2 mg/kg IV Atlas Alternative: vecuronium 0.1 mg/kg IV

Fentanyl administration Unit

Dilute the dose in 2 mL NS and infuse over 3 min, then flush with 1 mL NS over 3 min. Giving it slowly matters: fentanyl can cause chest wall rigidity.

Pre-ductal SpO₂ targets

Time after birth	Target SpO ₂
1 min 8th only	60–65%
2 min	65–70%
3 min	70–75%
4 min	75–80%
5 min	80–85%
10 min	85–95%

Values from 2 min on are the same in both editions. 9th dropped the 1-minute row.

Starting FiO₂ 9th

- ≥35 wk: 21%
- 32–34 wk: 21–30%
- <32 wk: ≥30%

9th A reasonable approach is to adjust FiO₂ in 20–30% steps every 30 s until the target is reached.

ETT tip and confirmation

Tip position on film

Target T1–T2 (optimal in recent trials).

Dir. Between the clavicles and carina, at least 0.5–1.0 cm above the carina. Carina at about T4. Her sheet lists T2–T3.

Head position moves the tube **Dir.**

Neck **flexion**: tube moves **down** (deeper).

Neck **extension**: tube moves **up**.

Confirm placement

- Rising heart rate
- Chest rise
- Symmetric breath sounds
- Exhaled CO₂ detector color change
- Chest x-ray

Airway

Airway

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ETT size

ID (mm)	8th	9th
2.0	—	Optional for <800 g (22–25 wk), if available
2.5	<1000 g <28 wk	<800 g (22–25 wk) 800–1200 g (26–28 wk)
3.0	1000–2000 g 28–34 wk	1201–2200 g 29–34 wk
3.5	>2000 g >34 wk	>2200 g >34 wk

Where depth is measured

8th Tip to **upper lip**.

9th Tip to **front edge of the upper gum, midline**.

Supraglottic airway **9th**

Can be used as the primary device if face-mask ventilation or intubation is unsuccessful or unavailable (≥34 wk).

Airway

Airway

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Emergency access and meds

Emergency UVC

Low-lying: insert **2–4 cm below the skin**, just until blood returns. IO is the alternative.

Epinephrine 0.1 mg/mL

IV / IO	0.01–0.03 mg/kg = 0.1–0.3 mL/kg Suggested start 0.02 mg/kg = 0.2 mL/kg Flush 3 mL NS
ETT	0.05–0.1 mg/kg = 0.5–1 mL/kg Only while IV access is being obtained
Repeat	Every 3–5 min

Volume

NS **10 mL/kg** over 5–10 min, via UVC or IO.

No change to epinephrine or volume dosing appeared in the 9th edition change summaries reviewed. 📌

Resus

Resus

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NRP 8th vs 9th

	8th	9th
Cord clamping	Defer 30–60 s	Defer ≥60 s
Cord milking <28 wk	—	Not recommended (severe IVH risk)
Initial FiO₂	≥35 wk: 21% <35 wk: 21–30%	≥35 wk: 21% 32–34 wk: 21–30% <32 wk: ≥30%
SpO₂ table starts	1 min	2 min
PPV rate	40–60/min	30–60/min
Initial PIP	20–25 cm H ₂ O 📌	≥32 wk: 25 (25–30) <32 wk: 20–25
PEEP	5	5
Initial steps	Includes "suction if needed"	Suction removed from the initial-steps box
Corrective steps	—	Start after 15–30 s of PPV
ETT	Depth to lip	Depth to gum; new size table

Resus

Resus

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Resus doses by weight

kg	IV/IO epi 0.2 mL/kg	ETT epi 1 mL/kg	NS 10 mL/kg
0.5	0.1	0.5	5
0.75	0.15	0.75	7.5
1	0.2	1	10
1.5	0.3	1.5	15
2	0.4	2	20
2.5	0.5	2.5	25
3	0.6	3	30
3.5	0.7	3.5	35
4	0.8	4	40

All volumes in mL. Epinephrine is the 0.1 mg/mL concentration. IV/IO column is the suggested starting dose; the range is 0.1–0.3 mL/kg. ETT column is the top of the 0.5–1 mL/kg range.

UVC

Tip position

IVC–right atrial junction, just above the diaphragm.

Dir. T8–T9. Follows the ductus venosus. Sits to the right of the spine on x-ray.

Insertion depth (cm), add stump length

Shukla	$[(3 \times \text{BW kg}) + 9] \div 2 + 1$ (half the high UAC + 1)
Revised	$[(3 \times \text{BW kg}) + 9] \div 2$ Dropping the +1 reduced over-insertion without more low lines
Surface	Umbilicus-to-nipple distance – 1 cm

Catheter [Atlas](#)

3.5 Fr <3.5 kg, 5 Fr >3.5 kg.

Confirm with AP and lateral films or ultrasound. Emergency UVC: see Resus section.

Removing a UAC

Steps [Dir.](#)

1. Stopcock off to the patient.
2. Stop IV fluid flow.
3. Pull back to the 2–3 cm mark on the catheter.
4. Wait 30–60 min.
5. Pull the catheter.

Reminders for every umbilical line

- Add the stump length to every formula.
- Confirm the tip on film before using the line.
- A formula is a starting point; the film decides.

X-ray tip targets

Device	Target
ETT	T1–T2 Dir. T2–T3; carina $\approx$ T4
UAC high	T6–T9 Dir. T6–T10
UAC low	L3–L4
UVC	IVC–RA junction Dir. T8–T9
PICC, arm or scalp	SVC–RA junction Dir. T3–T4
PICC, leg	IVC–RA junction; check a lateral

Where the director's sheet differs from current sources, both are shown. Unit policy decides.

Umbilical line depth by weight

BW kg	UAC Shukla	UAC Wright	UVC Shukla	UVC revised
0.5	10.5	9	6.3	5.3
0.75	11.3	10	6.6	5.6
1	12	11	7	6
1.25	12.8	12	7.4	6.4
1.5	13.5	13	7.8	6.8
2	15	—	8.5	7.5
2.5	16.5	—	9.3	8.3
3	18	—	10	9
3.5	19.5	—	10.8	9.8
4	21	—	11.5	10.5

Centimeters **before** adding stump length. Wright shown only to 1.5 kg because it was studied in VLBW infants. Rounded to 0.1 cm.

Lines

Doses, sizes, and sources

From MacDonald's Atlas Atlas

Rocuronium	0.6–1.2 mg/kg IV
Fentanyl	1–4 mcg/kg IV
UAC heparin	1–2 units/mL
UAC catheter	3.5 Fr <1200 g, 5 Fr >1200 g
UVC catheter	3.5 Fr <3.5 kg, 5 Fr >3.5 kg

Confirm doses with your unit's order set.

Sources

AAP NRP 9th ed launch communication (Oct 2025); CPS, NRP 9th ed changes for busy clinicians (2026); In[Neo]Sight ETT calculator (NRP 9th tables); PubMed 40739063 (ETT tip); NCCC umbilical line guidelines, mombaby.org (2022); Medscape UAC (2024); PubMed 23503982 (revised UVC); NCT02939690 (UN–1); Iran J Pediatr 2024, citing NANN (PICC); Pediatr Neonatol 2023, POCUS PICC; neonatal atropine RCT; VanLooy NICU premedication algorithm; PMC8228479 (emergency UVC); NRP textbook via tomwademd.net (volume); MacDonald's Atlas of Procedures in Neonatology, 6th ed (ch. 31, 32, 38); program director's NNP Brain; unit RSI poster.

Ref

UAC

Tip position

High (preferred): T6–T9, above the celiac axis. Dir. sheet lists T6–T10.

Low: L3–L4, above the aortic bifurcation (L4–L5). Low lines displace more easily and are not preferred.

Insertion depth (cm), add stump length

Shukla	(3 × BW kg) + 9
Wright	(4 × BW kg) + 7 Studied in VLBW infants

Precomputed depths are on the Lines depth table page.

Catheter and flush Atlas

3.5 Fr <1200 g, 5 Fr >1200 g. Heparin 1–2 units/mL in 0.45–0.9 NS.

Confirm tip position on film before use.

Lines

PICC

Measure (NANN)

Arm or scalp	Site → along the vein → right sternal border at the 3rd intercostal space. Arm at 90°.
Leg	Site → along the vein → just right of the umbilicus → xiphoid.

Tip position

SVC–RA or IVC–RA junction, outside the heart.

Dir. SVC tip at T3–T4.

Film tips

Dir. You can insert with the arm in any position. Before the x-ray, return the arm to the position the infant usually lies in.

A leg line can divert into the ascending lumbar vein and still look correct on an AP film. Get a lateral.

Lines